

Experiment No.3

Aim: To study and implement Bare-metal virtualization using Xen, HyperV or VMware Esxi.

Theory:

Bare-metal virtualization, also known as Type 1 or hypervisor-based virtualization, involves installing a virtualization layer directly on the physical hardware without the need for an underlying operating system. Xen, Hyper-V, and VMware ESXi are prominent examples of hypervisors that enable the creation and management of multiple virtual machines (VMs) on a single physical server. This architecture offers superior performance compared to hosted virtualization and is widely used in enterprise environments for its efficiency and resource optimization.

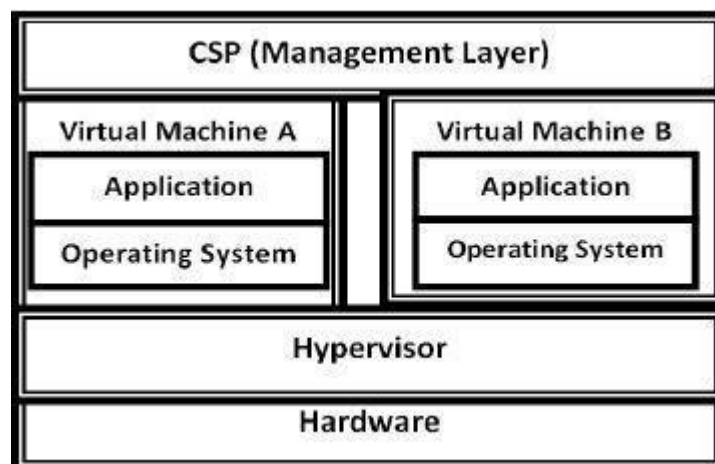


Figure 4 Bare metal virtualization architecture

At the Heart of Hardware:

Bare-metal virtualization offers a unique approach to resource management by placing the virtualization software, known as a hypervisor, directly on the physical hardware of a server. This eliminates the need for a host operating system, creating a streamlined environment for virtual machines (VMs) to reside.

The Hypervisor Takes Charge:

Acting as a conductor, the bare-metal hypervisor assumes control of the physical hardware's resources, including CPU, memory, storage, and networking. It then divides these resources into virtual pools, allowing for the creation and management of multiple independent virtual machines.

Creating Virtual Worlds:

Each VM operates as a self-contained computer, complete with its own virtual hardware, operating system, and applications. This isolation ensures that VMs run independently of one another, even as they share the underlying physical resources.

Managing the Virtual Landscape:

Specialized VM management tools provide a means to create, configure, monitor, and maintain virtual machines efficiently. These tools enable administrators to optimize resource allocation, troubleshoot issues, and ensure optimal performance across the virtual environment.

Benefits Unleashed:

Bare-metal virtualization offers several compelling advantages:

- **Performance Edge:** Direct hardware access often yields better performance compared to hosted virtualization approaches.
- **Security Fortification:** Isolating VMs from the underlying hardware can bolster security and reduce attack surfaces.
- **Efficiency in Action:** VMs can be rapidly provisioned, migrated, and managed, leading to enhanced resource utilization.
- **Scalability on Demand:** Additional VMs can be added or removed as needed, allowing adaptation to fluctuating workloads.
- **Cost Savings Realized:** Consolidating multiple workloads onto a single physical server can significantly reduce hardware expenses.

Common Bare-metal Hypervisors:

- VMware ESXi
- Microsoft Hyper-V
- Citrix XenServer
- KVM

Empowering Diverse Applications:

Bare-metal virtualization proves instrumental in a wide array of domains:

- Server consolidation
- Cloud computing
- Data center optimization
- High-performance computing (HPC)
- Virtual desktop infrastructure (VDI)
- Testing and development environments

By harnessing the power of bare-metal virtualization, organizations can unlock a path toward enhanced efficiency, flexibility, and cost savings within their IT infrastructure.

Installing Xen Center on Windows 10:

Step 1: Download Xen Center Installer

- Visit the Citrix XenServer Download page.
- Download the XenCenter installer suitable for Windows.

Step 2: Install XenCenter

- Run the downloaded Xen Center installer.
- Follow the on-screen instructions to complete the installation.

Step 3: Launch Xen Center

- Once installed, launch Xen Center on your Windows 10 machine.
- The first time you open XenCenter, you'll be prompted to connect to a server. Leave this empty for now.

Connecting Xen Center to Xen Server:

Step 4: Install XenServer on a Dedicated Server

- Acquire a dedicated server where XenServer can be installed. Download the XenServer ISO from the Citrix XenServer Download page.
- Install XenServer on the dedicated server by booting from the ISO and following the on-screen prompts.

Step 5: Configure Networking on XenServer

- Assign an IP address to your XenServer during the installation process.
- Ensure that the networking settings on XenServer are properly configured, including DNS resolution.

Step 6: Connect XenCenter to XenServer

- In XenCenter, click on "New Server" in the toolbar.
- Enter the IP address or hostname of your XenServer and provide the login credentials.
- Click "Connect."

Step 7: Manage Virtual Machines

- Once connected, you can start managing your XenServer from XenCenter.
- Create, start, stop, and manage virtual machines on the XenServer using the XenCenter interface.

Step 8: Install and Manage Virtual Machines

- Install and configure virtual machines on the XenServer using XenCenter.
- Utilize XenCenter features to monitor performance, allocate resources, and manage virtual machine snapshots.

Stepwise Procedure to install Ubuntu Server:

Step 1: Open XenCenter

- Launch XenCenter on your Windows machine.

Step 2: Create a New VM

- Click on the "New VM" button in the XenCenter toolbar.
- In the "New VM" wizard, select "Ubuntu" as the OS type.

Step 3: Provide VM Details

- Enter a name for your virtual machine
- Choose the appropriate storage repository for the VM.
- Assign an amount of memory (RAM) to the VM.
- Select the number of virtual CPUs for the VM.

Step 4: Attach Ubuntu ISO

- In the "Installation Media" step of the wizard, choose "Use an ISO library or physical drive."
- Click on the "Add" button and browse to the location where you downloaded the Ubuntu Server ISO. Select the ISO file and click "OK."

Step 5: Networking Configuration

- In the "Networking" step, configure the network settings for your VM. Choose the appropriate network and assign an IP address if needed.

Step 6: Storage Configuration

- Choose the storage repository where the VM's virtual disk will be stored.
- Specify the size for the virtual disk.

Step 7: Review and Create VM

- Review the VM configuration in the summary.
- Click "Create Now" to create the VM.

Step 8: Start the VM

- With the VM selected, click on the "Start" button in the toolbar.
- The VM will boot from the Ubuntu Server ISO, initiating the installation process.

Step 9: Install Ubuntu Server

- Follow the on-screen instructions to install Ubuntu Server.
- Configure language, keyboard layout, and network settings as needed.
- When prompted, choose the disk where you want to install Ubuntu.

Step 10: Complete Installation

- Complete the Ubuntu Server installation by setting up a user account and password.
- Once the installation is complete, the VM will reboot.

Step 11: Eject ISO

- In XenCenter, right-click on the VM and choose "Eject CD/DVD."
- This ensures that the VM boots from the virtual hard drive instead of the ISO.

Step 1: Install Xen Server

1. Insert Bootable Xen Server CD into CDROM and make first boot device as a CDROM from BIOS



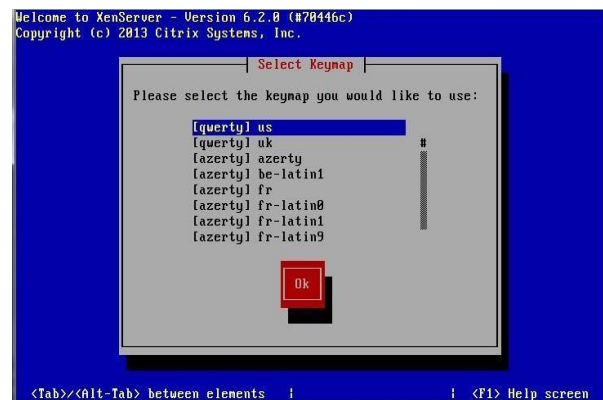
2. Press F2 to see the advanced options otherwise press Enter to start installation



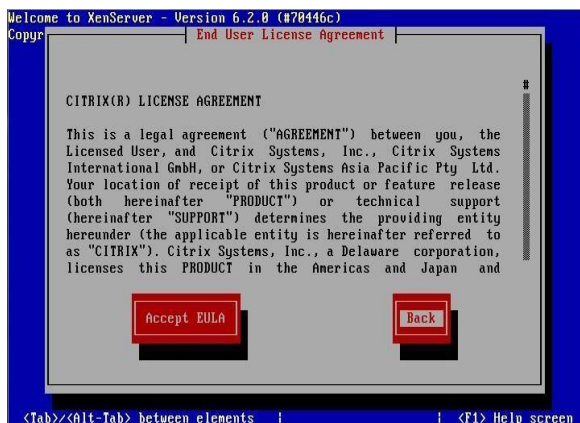
3. Select Keyboard Layout



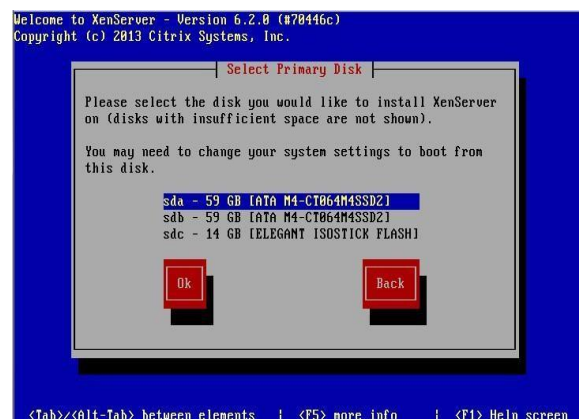
4. Press Enter to load Device Drivers



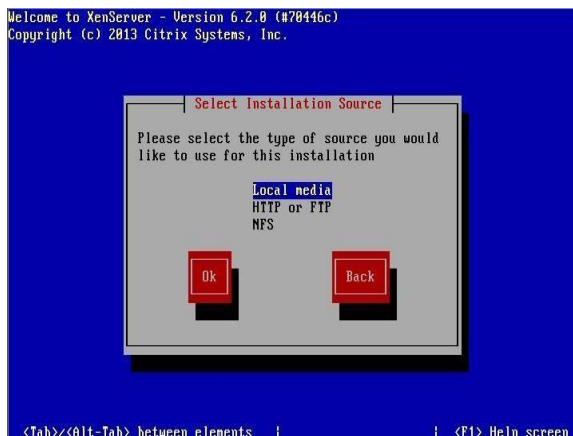
5. Press Enter to Accept End User License Agreement



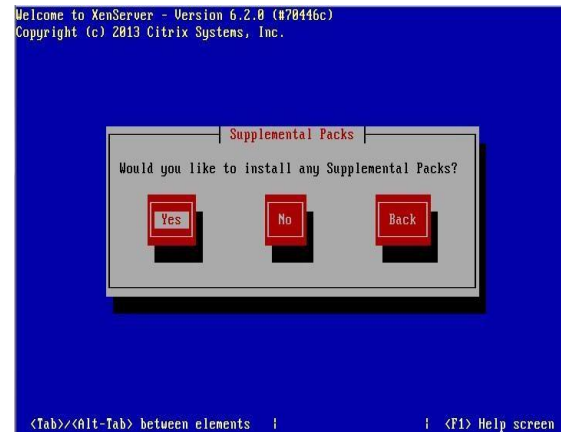
6. Select appropriate disk on which you want to install Xen Server



7. Select Appropriate installation Media



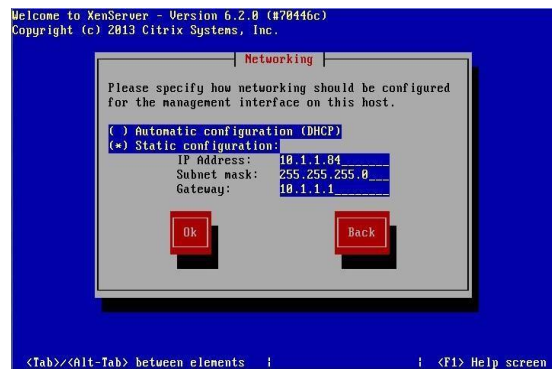
8. Select Additional Packages for installation



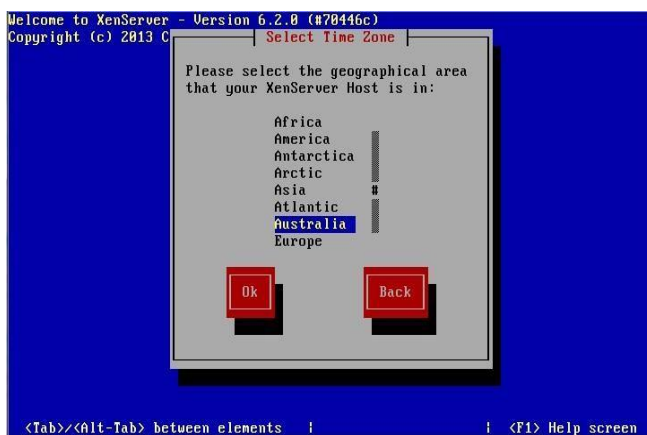
9. Specify Root Password



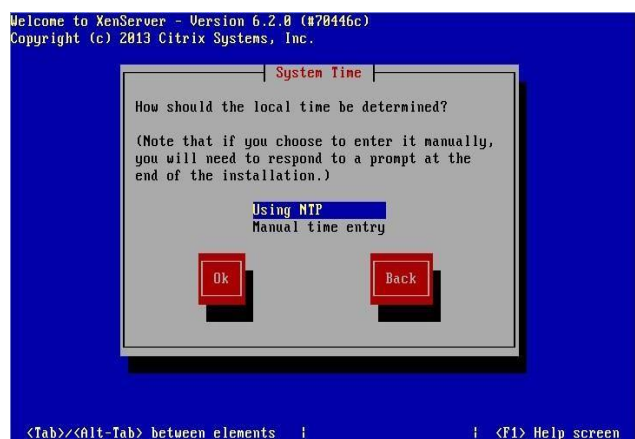
10. Specify IP Address to a Xen Server



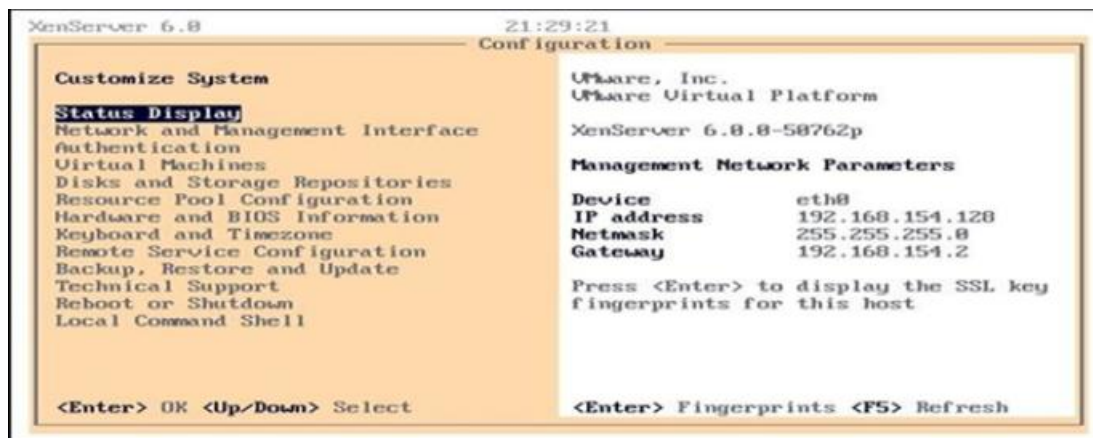
11. Select Time Zone



12. Specify NTP Servers address or use manual time entry the start installation



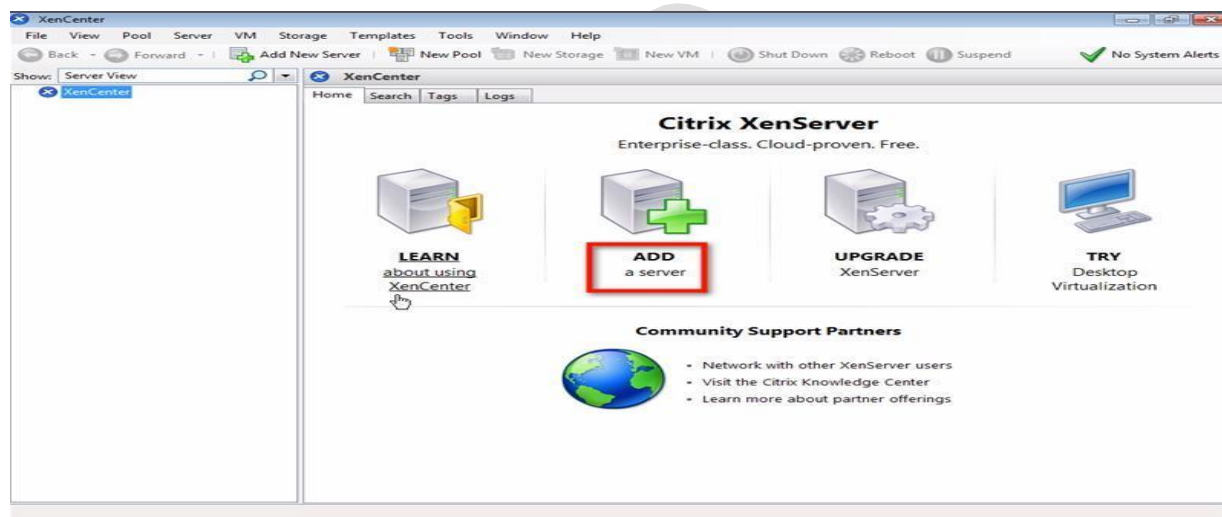
Once Installation is done you will see the final screen shown below



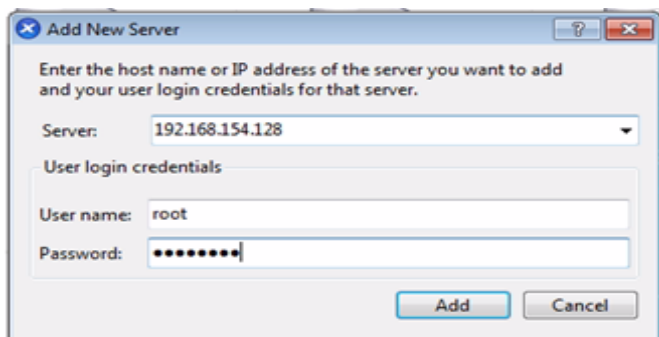
Xen Server Final Screenshot

Step 2: Connect Xen Server to Xen Center

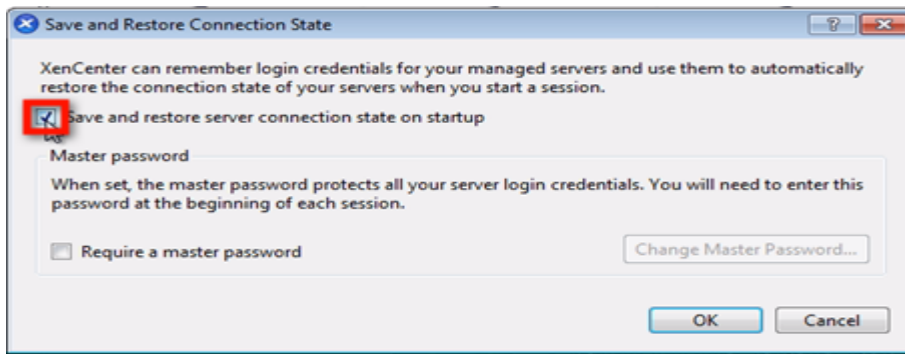
Firstly, download the xen center management utility from xen server by opening the xen servers IP address as a URL on browser. Once Xen center is downloaded, install it. Open Xen center from start menu of Windows. Then to connect to the Xen Server host you configured earlier, click Add a server.



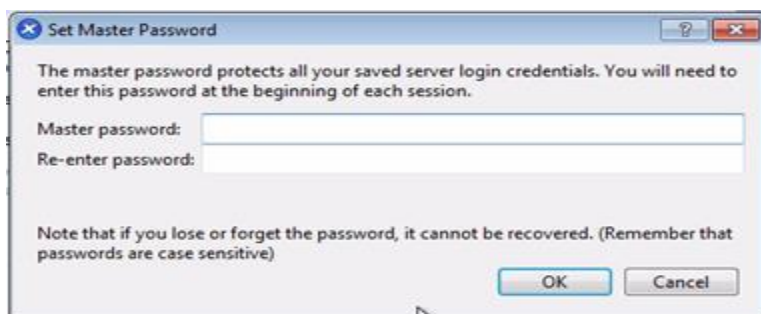
Enter the IP address I asked you to take note of earlier. Also enter the password you assigned for your root account. Click Add



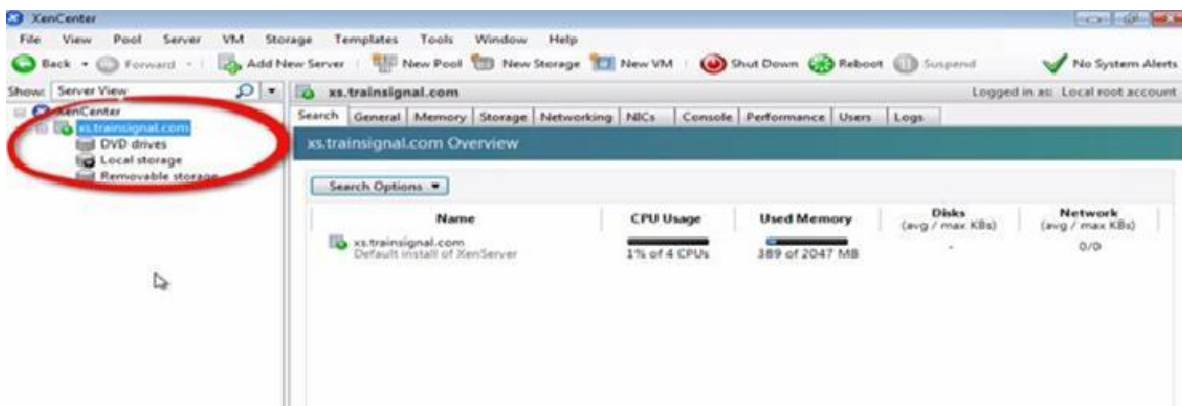
One of the first things you want to make sure as you're adding a new Xen Server to Xen Center is to save and restore the server connection state on startup. Check the box that will do just that.



Once you do that, you will be allowed to configure a master password for all the Xen Servers you'll be associating with this Xen Center. Click the require a master password checkbox if that's what you want to do, and then enter your desired master password in the fields provided.

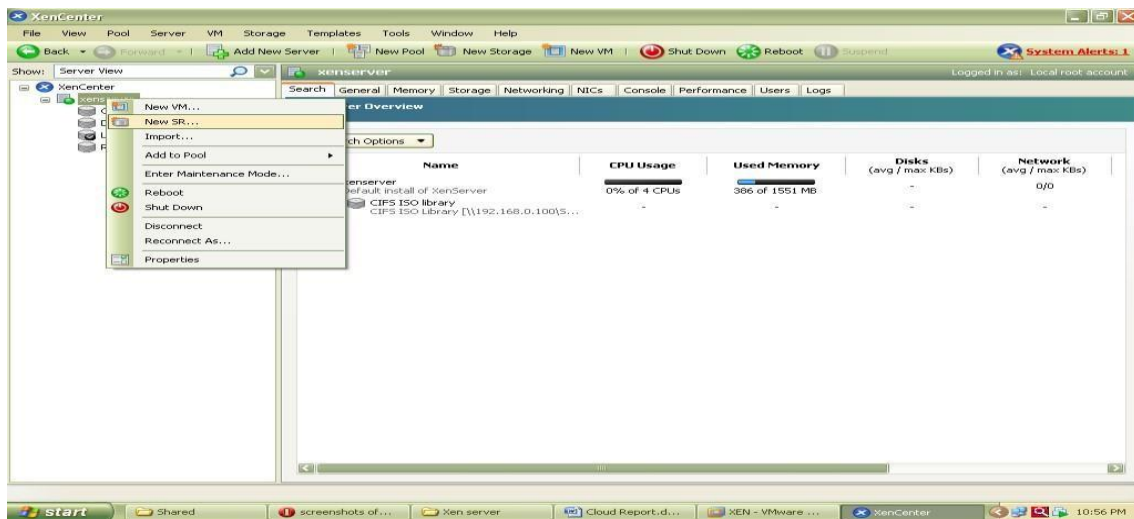


After that Click OK, you'll be brought back to the main screen, where you'll see your Xen Server already added to Xen Center.

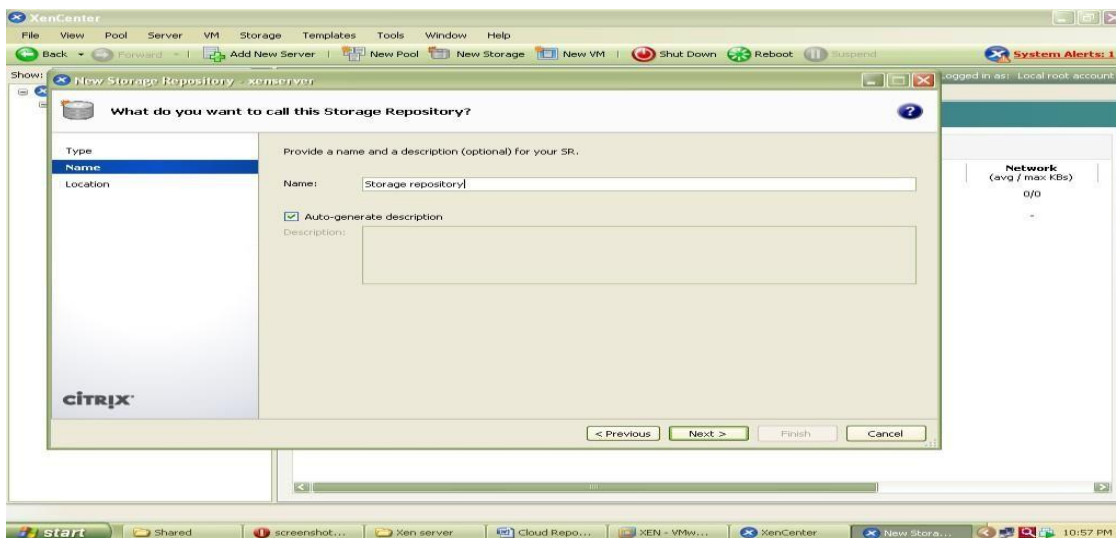


Step 3: Create Storage Repository and installing VM

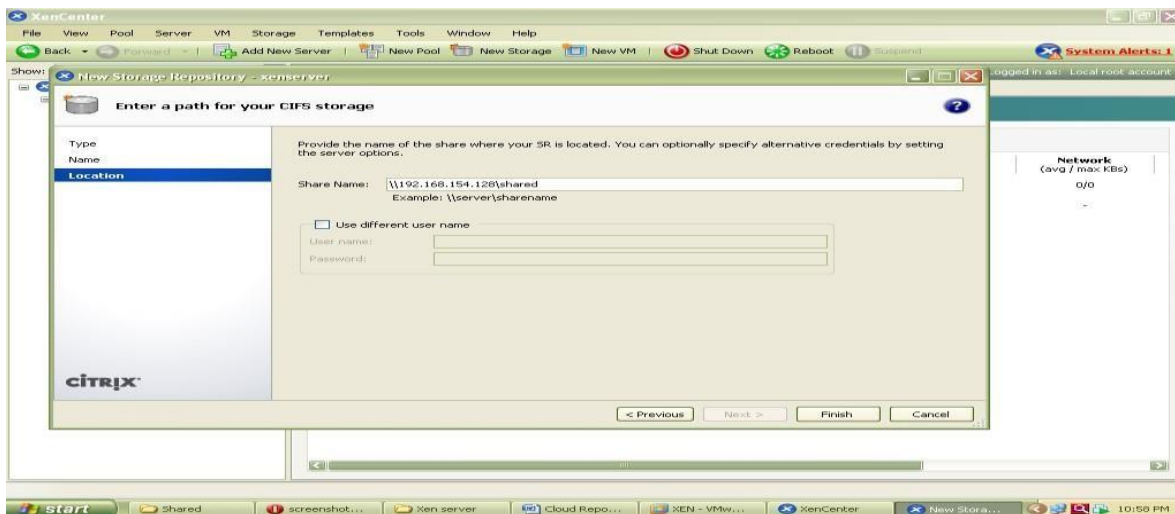
Now before creating VM, we have to create storage repository first which is nothing but shared directory on Xen Center which holds all iso files and which is required to install operating system on Xen Server its steps are as follows. Right click on Xen Server icon on Xen Center and click on New SR



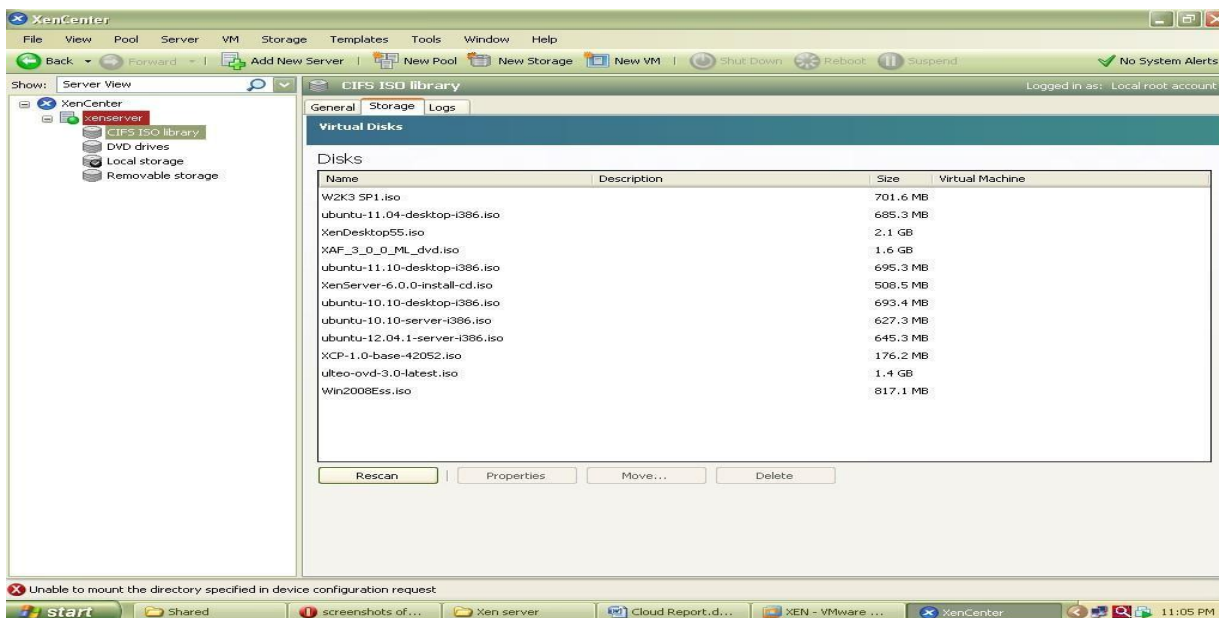
Now Select Windows CIFS library then Specify Storage Repository Name



Now specify path of shared folder at client side which holds all iso files of OS or VM which we are going to install on Xen Server.

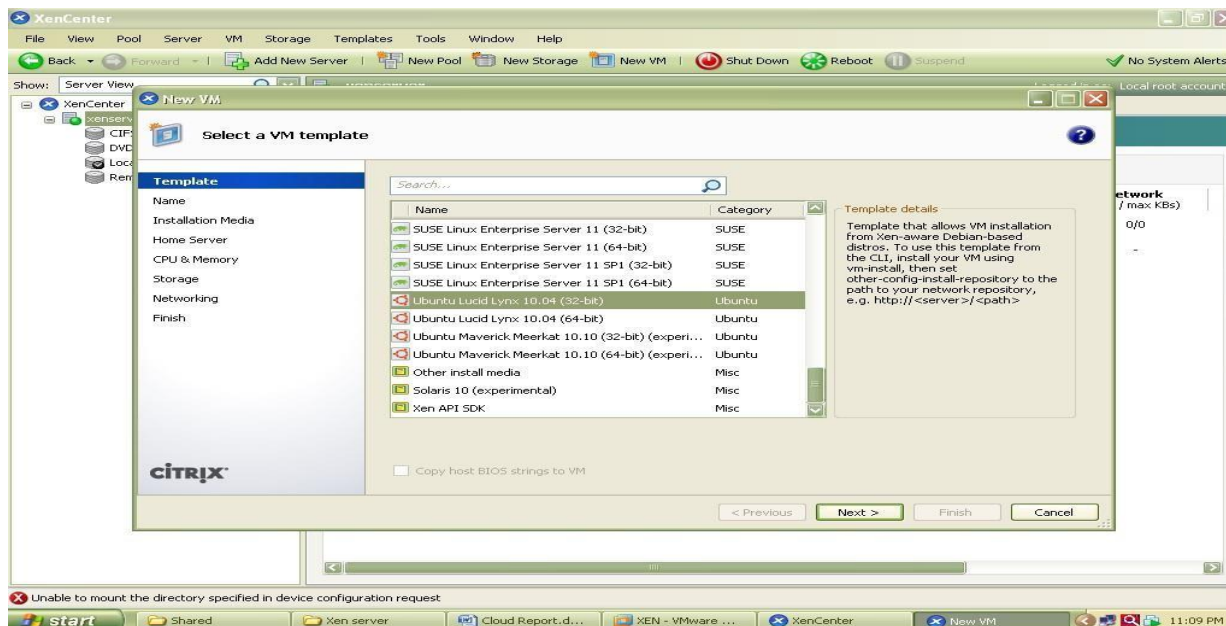


At the end Click on finish to create SR. To check all iso files click on CIFS library and select storage this will show you all iso files.

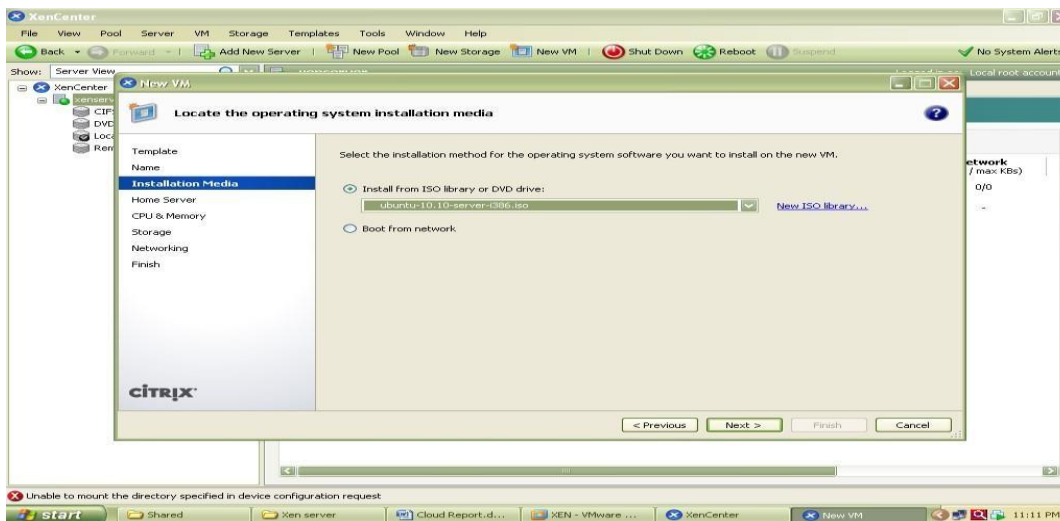


Installation of UBUNTU Server on Xen Server

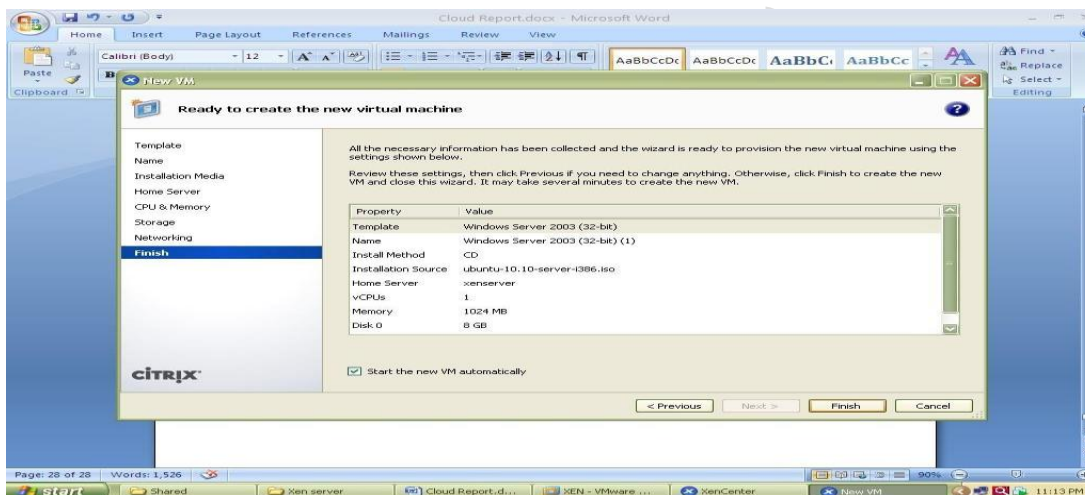
Step1: Right click on Xen Server icon on Xen Center and select New VM then select Operating System to be install here select Ubuntu Lucid Lynx and click on next



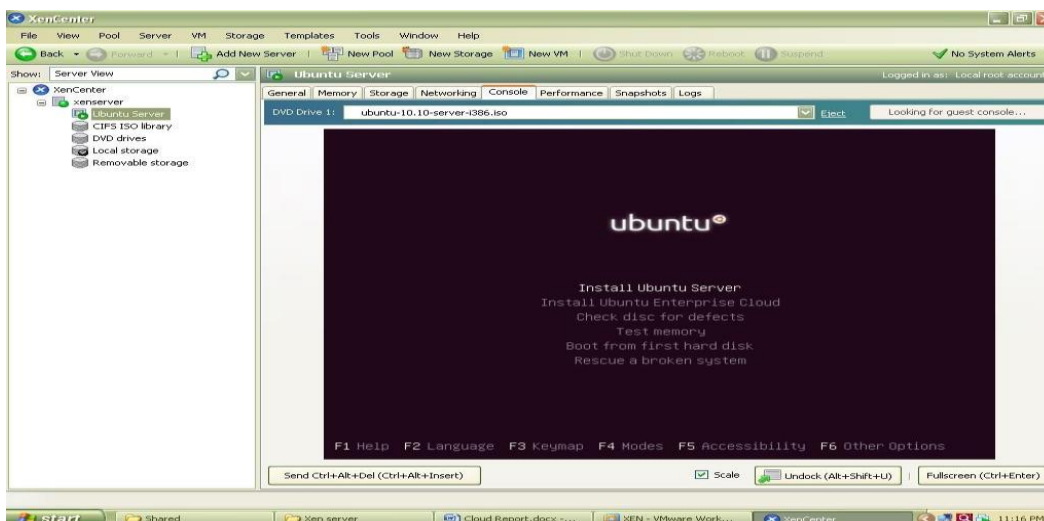
Step2: Now specify Instance Name as ubuntu Server then select iso file of Ubuntu server 10.10 to be install



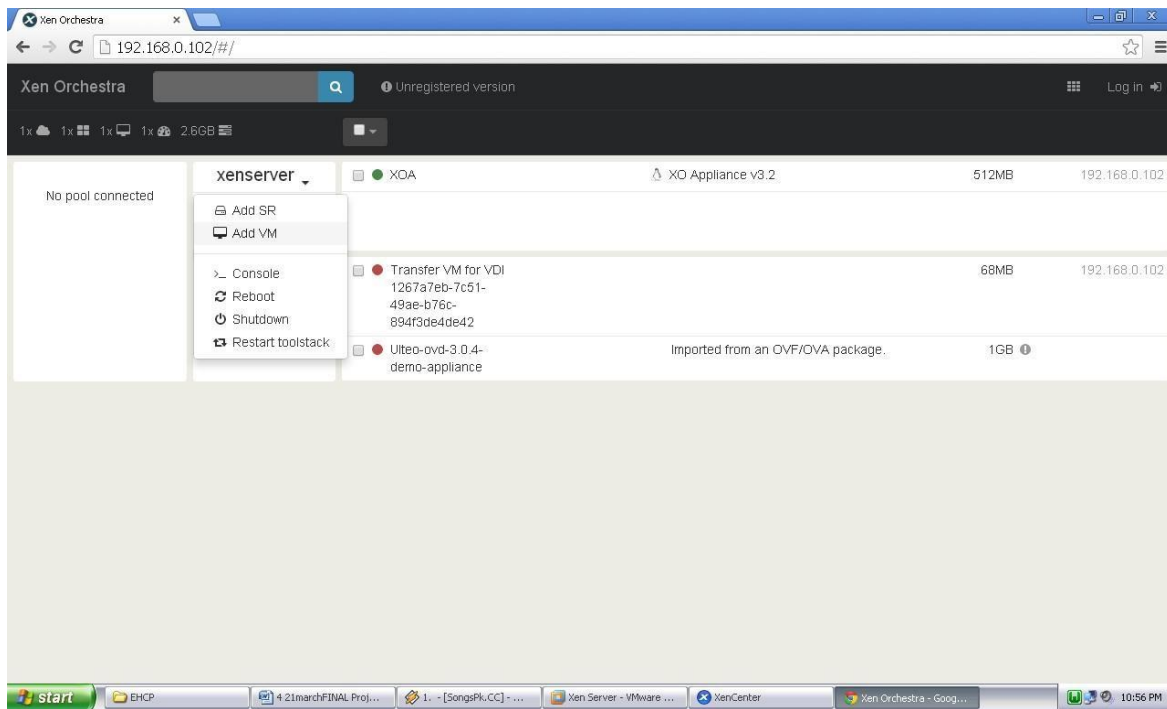
Step3: Now select hardware for VM i.e. **no. of CPU's and memory** after that select → **local storage** → **network** and then click on **finish**.



Step4: Now go to console tab to install ubuntu and follow installation steps.



The Xen Orchestra provides web-based functionality of Xen Center. It provides access to all the VMs with their lifecycle management which are installed over Xen Server shown in figure



Xen Orchestra (XOA) Portal

Conclusion: Thus, we have successfully demonstrated the installation of Xen Server and subsequently installed and configured Ubuntu system.